

5.1.1 Model 1: Linear Regression (Baseline)

Baseline model per assignment spec requirement. Trained on `X_train_scaled.csv` (log1p-transformed numeric features, per Section 3.11.2) since Linear Regression is scale-sensitive.

```
In [ ]: import pandas as pd
from sklearn.linear_model import LinearRegression
import sys, os
sys.path.append(os.path.dirname(os.path.abspath('__file__')))
from model_utils import evaluate_model, cross_validate_model

MODELLING_DIR = os.path.join("modelling_data")

X_train = pd.read_csv(os.path.join(MODELLING_DIR, "X_train_scaled.csv"))
X_test = pd.read_csv(os.path.join(MODELLING_DIR, "X_test_scaled.csv"))
y_train = pd.read_csv(os.path.join(MODELLING_DIR, "y_train.csv"))["price"]
y_test = pd.read_csv(os.path.join(MODELLING_DIR, "y_test.csv"))["price"]

print(f"X_train: {X_train.shape} | X_test: {X_test.shape}")

X_train: (3004, 51) | X_test: (751, 51)
```

Train

```
In [ ]: model = LinearRegression()
model.fit(X_train, y_train)
print("Model trained.")

Model trained.
```

Evaluate

Metrics computed on both train and test sets — the gap between them is needed for Section 6.2's overfitting/underfitting analysis.

```
In [ ]: train_metrics = evaluate_model(model, X_train, y_train, label="Train")
print()
test_metrics = evaluate_model(model, X_test, y_test, label="Test")

Train RMSE:  RM 168,260  (48.1% of median price)
Train MAE:   RM 86,459
Train MAPE:  21.2%
Train R2:    0.7366

Test RMSE:   RM 194,955  (54.2% of median price)
Test MAE:    RM 92,474
Test MAPE:   19.8%
Test R2:     0.6547
```

5-fold Cross-Validation

Run on `X_train` only (`X_test` stays untouched) to get a more robust performance estimate than a single train/test split, per Section 5.2's cross-validation requirement.

```
In [ ]: cv_results = cross_validate_model(LinearRegression(), X_train, y_train, n_splits=5)

5-fold CV (mean +/- std):
RMSE:  RM 171,190 +/- 26,902  (48.7% of median price)
MAE:   RM 88,637 +/- 5,162
MAPE:  21.8% +/- 1.1%
R2:    0.7214 +/- 0.0486
```

Save Trained Model

Saved for the Streamlit prototype (Section 8) to load directly, without retraining.

```
In [ ]: import joblib
MODEL_DIR = os.path.join("..", "streamlit")
os.makedirs(MODEL_DIR, exist_ok=True)
model_path = os.path.join(MODEL_DIR, "linear_model.pkl")
joblib.dump(model, model_path)
print(f"Model saved to {model_path}")

Model saved to ..\streamlit\linear_model.pkl
```